

- 1 (a) The mammalian kidney is an organ involved in homeostasis.

Explain what is meant by the term *homeostasis*.

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..... [1]

- (b) Fig. 1.1 shows a section through a kidney.

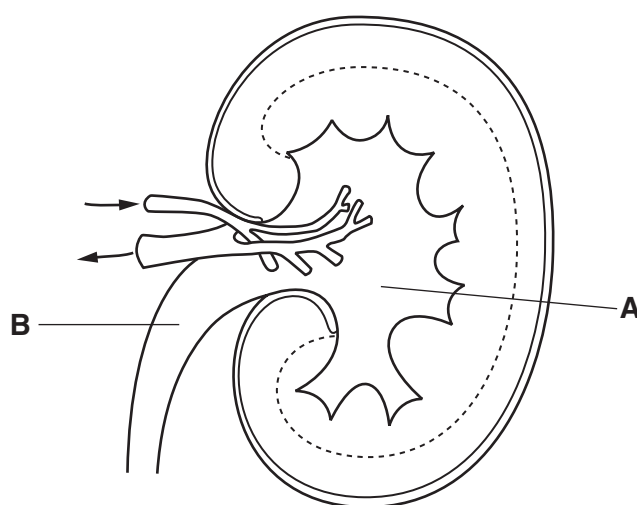


Fig. 1.1

- (i) With reference to Fig. 1.1, name structures **A** and **B**.

A

B

[2]

- (ii) On Fig. 1.1, use label lines and letters to label where:

U – ultrafiltration occurs

L – the loop of Henle is found

C – blood urea concentration is low.

[3]

(c) Describe the roles of the hypothalamus and the posterior pituitary in osmoregulation.

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[Total: 11]